

Public Lighting Modernization Project

Project Overview

The Public Lighting Modernization Project is a strategic initiative for Willow Creek. It will replace obsolete lighting infrastructure with an intelligent, sustainable LED system. Launched on October 15, 2023 with a budget of \$300,000, the project supports the town's broader ecological transition and quality-of-life strategy.

Context and Rationale

Willow Creek's public lighting system currently presents several urgent issues:

- some installations are more than 30 years old;
- excessive energy consumption, representing about 35% of the municipal electricity bill;
- recurring outages that create areas of perceived insecurity;
- lighting that does not match the real needs of different urban areas;
- light pollution affecting nighttime biodiversity;
- maintenance costs that continue to rise.

This project complements other municipal initiatives, including central road renovation, the development of municipal internet services, and the urban surveillance system. Together, these projects create useful synergies across the town's infrastructure.

Project Objectives

- Reduce energy consumption by switching to latest-generation LED technologies.
- Lower operating costs through more efficient and durable equipment.
- Improve lighting quality to strengthen resident safety.
- Limit light pollution through intelligent light-flow management.
- Highlight architectural heritage with appropriate lighting.
- Create a connected lighting network that can be remotely managed and programmed.
- Develop a resilient system for outages and extreme weather conditions.

Scope of Work

The project covers the entire town, with priority given to the following areas:

- historic downtown: 250 light points;
- main traffic corridors: 180 light points;
- areas around schools: 75 light points;
- priority residential areas: 220 light points;
- public parks and gardens: 95 light points.

In total, 820 light points will be replaced or modernized, representing about 65% of the town's public-lighting stock.

Technical Description of the Work

Initial Assessment

- complete audit of existing equipment with GIS mapping;
- photometric measurements in all covered areas;
- analysis of pole and power-network aging;
- environmental and energy-impact study;
- modeling of new neighborhood-level needs.

Equipment Replacement

- removal of old high-pressure sodium and mercury-vapor fixtures;
- installation of high-efficiency LED luminaires above 140 lm/W;
- installation of intelligent controllers on each light point;
- replacement of obsolete electrical cabinets;
- renovation of poles showing corrosion or instability;
- upgrading underground power networks to current standards.

Intelligent Management System

- deployment of a wireless mesh network for communication between luminaires;
- installation of a centralized remote-management system;
- use of presence sensors in low-traffic areas;
- integration of photocells to adapt to ambient light;
- development of a control interface for public works teams.

Technical Specifications

- LED luminaires with a color rendering index (CRI) above 80;
- color temperature adapted to use: 3000K in residential areas and 4000K on road corridors;
- minimum IP66 protection rating and IK08 impact resistance;
- minimum service life of 100,000 hours (L80B10);
- 10-year manufacturer warranty on all equipment;
- integrated surge protection devices.

Innovation and Sustainable Approach

- automatic dimming based on time and use;
- smart presence detection with a light wave that follows pedestrians;
- hybrid grid/solar supply for selected strategic points;
- adaptive lighting based on weather conditions;
- integrated environmental sensors for air quality, noise, and traffic;
- anti-glare solutions to protect nocturnal wildlife.

Phasing and Preliminary Timeline

The deployment will follow five phases:

Phase 1 — October to November 2023

- in-depth assessment and complete mapping;
- equipment orders and logistics preparation.

Phase 2 — December 2023 to January 2024

- downtown and main corridors;
- deployment of the centralized management system.

Phase 3 — February to March 2024

- areas around schools;
- priority residential areas in the north sector.

Phase 4 — April to May 2024

- priority residential areas in the south sector;
- public parks and gardens.

Phase 5 — June 2024

- testing, adjustments, and system optimization;
- training for municipal staff.

Final acceptance of the project is planned for the end of June 2024, for a total duration of eight months.

Detailed Budget

- preliminary studies and engineering: \$21,000 (7%);
- LED luminaires: \$165,000 (55%);
- intelligent management system: \$45,000 (15%);
- labor and installation: \$51,000 (17%);
- pole and network renovation: \$15,000 (5%);
- contingency and miscellaneous: \$3,000 (1%).

Financial Analysis and Return on Investment

The investment will be offset by savings generated by the modernization:

- current electricity consumption: 850 MWh/year;
- projected electricity consumption: 255 MWh/year;
- annual savings: 595 MWh, or about \$89,250 at the current rate;
- estimated maintenance-cost reduction: \$15,000 per year;
- calculated return on investment: 2.9 years;
- cumulative savings over 15 years: \$1.56 million.

Organization and Governance

The project will be led by the Energy and Sustainable Development team in collaboration with Public Works, a specialized public-lighting engineering firm, and the County Energy Partnership as a funding partner.

A technical committee will meet twice a month to track progress. It will include the energy manager, the deputy mayor for sustainability, the director of public works, a public safety representative, a biodiversity expert, and a representative of the local astronomy association.

Communication and Awareness Plan

- neighborhood public meetings before each phase of work;
- explanatory signage at work sites;
- articles in the municipal newsletter and on willowcreek.gov;
- mobile-app updates on areas currently being modernized;
- school activities on energy savings;
- guided night walks after renovation to show improvements.

Environmental Impact

- 70% reduction in electricity consumption linked to lighting;
- annual carbon-footprint reduction of 47 tons of CO₂;
- 65% reduction in light spill and light pollution;
- creation of dark corridors favorable to biodiversity.

Monitoring and Evaluation Indicators

Project effectiveness will be measured through:

- monthly electricity-consumption tracking;
- before-and-after light-pollution mapping;
- number of maintenance interventions;
- resident satisfaction surveys;
- local biodiversity impact measurements;
- lighting network availability rate.

Risk Management

Several risks have been identified, along with mitigation measures:

- supply difficulties: early ordering and multiple suppliers;
- compatibility with existing infrastructure: detailed preliminary diagnostics;
- public acceptance: information campaign and testing period;
- adverse weather: seasonal planning;
- cybersecurity threats to the smart system: reinforced security protocols.

Future Development

- extension to the remaining 35% of the lighting stock by 2026;
- integration with the broader smart-town strategy;
- development of connected services through the lighting network;
- testing photoluminescent road surfaces;
- using collected data to optimize other municipal services.

Impact on Residents' Daily Lives

- safer nighttime travel;
- better sleep quality through reduced light pollution;
- improved urban atmosphere and attractiveness of public spaces;
- enhanced visibility of architectural heritage;
- a stronger sense of safety in public spaces;
- renewed access to the night sky in an urban setting.

Conclusion

The Public Lighting Modernization Project is more than a technical upgrade. It is a deep transformation of the town's nighttime environment, balancing safety, energy savings, environmental protection, and technological innovation.

With its comprehensive and sustainable design, the project will embody Willow Creek's vision of a responsible, future-ready town while generating immediate and measurable benefits for all residents.